# Helix Forge Research Latency Quality Methods Study

HELIX FORGE LAB / p04-full-004689 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded latency quality evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000003 (research/literature-notes/2026-07-13/helix-forge-research/metric-contracts/tables/helix-forge-research-metric-contracts-metrics-v01.csv).

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 214 | 0.871 | 0.881 | 0.876 |
| slice-02 | 224 | 0.888 | 0.857 | 0.872 |
| slice-03 | 200 | 0.870 | 0.870 | 0.870 |
| slice-04 | 254 | 0.887 | 0.883 | 0.885 |
| slice-05 | 252 | 0.869 | 0.859 | 0.864 |